M101 · Unit 10 · Quick reference

Integral models, signs and dimensions
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Recall **Signed and geometric totals**	$I = \int_a^b f(x)\,dx$; $A = \int_a^b \lvert f(x) \rvert\,dx$. Forward bounds $a < b$.
Recall **Between boundaries**	Area $\int_a^b \lvert f - g \rvert\,dx$. Upper minus lower on each ordering interval.
Understand **Where to split**	Find zeros/sign changes of the integrand or boundary difference. A touching zero need not reverse sign.
Recall **Motion along a line**	$\Delta s = \int_a^b v\,dt$; $D = \int_a^b \lvert v \rvert\,dt$; $s(b) = s(a) + \Delta s$.
Recall **Interval mean**	$\overline{f} = \frac{1}{b-a}\int_a^b f\,dx$, $a < b$. Same units as f.
Understand **Weighted means**	$\overline{f} = (m_1L_1 + m_2L_2) / (L_1 + L_2)$. Weight by interval length.

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Recall **Stored amount**	$q = q_{\text{in}} - q_{\text{out}}$; $V(t) = V(a) + \int_a^t q(u)\,du$. Check empty/capacity boundaries.
Recall **Slice volume**	$V = \int_a^b A(x)\,dx$; $A \geq 0$. Circle: $A = \pi r^2$; square: $A = s^2$.
Consult **Supplied work model**	$W = \int_a^b F(x)\,dx$, signed force component in the chosen axis; $\mathrm{N\,m} = \mathrm{J}$.
Understand **Checks**	$D \geq \lvert \Delta s \rvert$; $m \leq \overline{f} \leq M$ if $m \leq f \leq M$. Check units, derivative, initial value and bounds.
Consult **Conditions and notation**	Continuous functions on bounded intervals; split piecewise continuous models at jumps. Mean requires positive interval length. Numerical variable units must match stated formula and bounds.

Consult · Python

```
def change(F, a, b):
    return F(b) - F(a)

def mean(F, a, b):
    if b <= a:
        raise ValueError("Require a < b")
    return change(F, a, b)/(b - a)

# Example: v(t) = 2*t - 2, reversal at t = 1
def F(t):
    return t*t - 2*t
D = abs(change(F, 0, 1)) + abs(change(F, 1, 2)) # 2
```

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